Ex. 8.8

<u>Q67</u> $\displaystyle\int_{-\infty}^{\infty} \frac{dx}{\sqrt{x^4+1}} = \int_{-\infty}^{-1} \frac{dx}{\sqrt{x^4+1}} + \int_{-1}^{1} \frac{dx}{\sqrt{x^4+1}} + \int_{1}^{\infty} \frac{dx}{\sqrt{x^4+1}}$

First integral:

$$\int_{-\infty}^{-1} \frac{dx}{\sqrt{x^4+1}} = \int_{1}^{\infty} \frac{du}{\sqrt{u^4+1}}$$

Set $u = -x \Rightarrow du = -dx$

when $x = -1$, $u = 1$

$x \to -\infty$, $u \to \infty$.

$$\frac{1}{\sqrt{u^4+1}} \sim \frac{1}{\sqrt{u^4}} = \boxed{\frac{1}{u^2}} \to \text{A candidate for LCT}$$

Let $f(u) = \dfrac{1}{\sqrt{u^4+1}}$, $g(u) = \dfrac{1}{u^2}$ for $u \geq 1$.

$$\lim_{u\to\infty} \frac{f(u)}{g(u)} = \lim_{u\to\infty} \frac{\frac{1}{\sqrt{u^4+1}}}{\frac{1}{u^2}} = \lim_{u\to\infty} \frac{u^2}{\sqrt{u^4+1}} = \lim_{u\to\infty} \frac{1}{\sqrt{1+\frac{1}{u^4}}} = 1 > 0$$

Since $\int_1^\infty \frac{1}{u^2}\,du$ converges, the integral $\int_1^\infty \frac{du}{\sqrt{u^4+1}}$ converges by LCT.

Second integral:

$$\int_{-1}^{1} \frac{dx}{\sqrt{x^4+1}}$$

Note that $x^4+1 \geq 1$ for $-1 \leq x \leq 1$

$$\Rightarrow \sqrt{x^4+1} \geq 1$$

$$\Rightarrow 1 \geq \frac{1}{\sqrt{x^4+1}} > 0 \quad - (\star)$$

$$\Rightarrow 0 \leq \int_{-1}^{1} \frac{dx}{\sqrt{x^4+1}} \underset{\text{valid because of } (\star)}{\leq} \int_{-1}^{1} 1 \, dx = 2$$

$\therefore \int_{-1}^{1} \frac{dx}{\sqrt{x^4+1}}$ converges.

Alternatively, one can just say $\int_{-1}^{1} \frac{dx}{\sqrt{x^4+1}}$ converges because it is a definite integral (the function contains no singular points)

Third integral: $\int_{1}^{\infty} \frac{dx}{\sqrt{x^4+1}}$ converges ~~from~~ (shown from the first integral)

$\therefore \int_{-\infty}^{\infty} \frac{dx}{\sqrt{x^4+1}}$ converges (since three integrals converge)

<u>Remark</u> ① $\int_{-\infty}^{\infty} f(x)\,dx \neq \underline{\lim_{c\to\infty} \int_{-c}^{c} f(x)\,dx}$

Improper integrals are not defined in this way!

② If we write $\int_{-\infty}^{\infty} \frac{dx}{\sqrt{x^4+1}} = \int_{0}^{\infty} \frac{dx}{\sqrt{x^4+1}} + \int_{-\infty}^{0} \frac{dx}{\sqrt{x^4+1}}$

and do the comparison with $g(x) = \frac{1}{x^2}$, we will get

$\int_0^{\infty} g(x)\,dx$ diverges.

In fact, it fails to satisfy the hypothesis of LCT:

we need to ensure $g(x)$ is continuous on the interval $\underline{[0,\infty)}$.

Here $g(x)$ is not continuous at $x=0$.

$[N,\infty)$ when consider $\int_N^{\infty} f(x)\,dx$

Q68 $\displaystyle\int_{-\infty}^{\infty} \frac{dx}{e^x + e^{-x}} = \int_{-\infty}^{0} \frac{dx}{e^x + e^{-x}} + \int_{0}^{\infty} \frac{dx}{e^x + e^{-x}}$

Note that $\dfrac{1}{e^x + e^{-x}} < \dfrac{1}{e^x}$ for $x \geq 0$

$$\int_0^\infty \frac{1}{e^x}\,dx = \lim_{b\to\infty} \int_0^b \frac{1}{e^x}\,dx = \lim_{b\to\infty} \left[-e^{-x}\right]_0^b$$

$$= \lim_{b\to\infty} \left[-e^{-b} + 1\right]$$

$$= 1$$

$\therefore \displaystyle\int_0^\infty \frac{1}{e^x}\,dx$ converges ~~by~~

$\therefore \displaystyle\int_0^\infty \frac{dx}{e^x + e^{-x}}$ converges by Direct Comparison Test

Note that $\displaystyle\int_{-\infty}^{0} \frac{dx}{e^x + e^{-x}} = \int_0^\infty \frac{du}{e^u + e^{-u}}$ $[u = -x] \Rightarrow \displaystyle\int_{-\infty}^{\infty} \frac{dx}{e^x + e^{-x}}$ converges.

(converges)